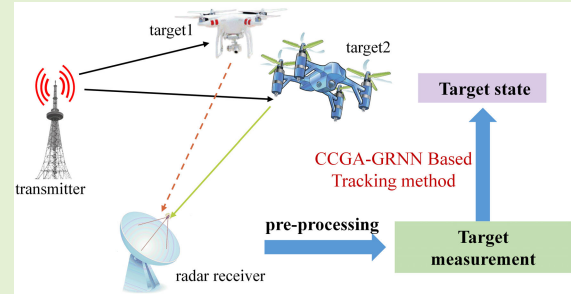


Target Tracking Algorithm Based on Generalized Regression Neural Network for Passive Bistatic Radar

Baoxiong Xu, Jianxin Yi[✉], *Member, IEEE*, Xianrong Wan[✉], and Feng Cheng[✉]

Abstract—In passive bistatic radar, high-performance target tracking is of crucial importance to a variety of applications. However, most existing tracking algorithms cannot achieve satisfactory performance due to poor azimuth estimation and the nonlinear relationship between the target position and measurements. This article proposes a novel tracking algorithm based on improved generalized regression neural networks (GRNNs) and Kalman filtering (KF). First, the dependency between the measurements and the target position is modeled by GRNN regression. A cooperative co-evolutionary genetic algorithm (CCGA) is then introduced to optimize the sample size and smoothing factor of the GRNN model to improve the performance. Finally, this article applies KF to CCGA-GRNN localization results and forms the smoothing trajectories of the targets. Simulation and experimental results show that the performance of the proposed algorithm is better than that of the geometric localization (GL), GRNN, and extreme learning machine (ELM)-based tracking approaches. Furthermore, field experiment data verify the feasibility of the proposed algorithm.

Index Terms—Co-evolutionary genetic algorithm (CCGA), generalized regression neural network (GRNN), Kalman filtering (KF), passive bistatic radar, target tracking.



ABBREVIATIONS

The following abbreviations are used in this article.

BTT	Bistatic target tracking.
CA	Constant acceleration.
CCGA	Cooperative co-evolutionary genetic algorithm.
ELM	Extreme learning machine.
GL	Geometric localization.
GRNN	Generalized regression neural network.
GPS	Global Positioning System.
IOs	Illuminators of opportunity.
KF	Kalman filtering.
MC	Monte Carlo.

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MRE	Mean relative error.
NN	Neural network.
PW	Parzen windows.
RBF	Radial basis function neural network.
RMSE	Root mean square error.
TDOA	Time difference of arrival.
2-D	Two-dimensional.
UAV	Unmanned aerial vehicle.
s_r	Receiver position.
s_t	Transmitter position.
x_k	Target state vector at time t_k .
z_k	Target measurement vector at time t_k .
$h(\cdot)$	Nonlinear measurement function.
ω_k	Measurement noise at time t_k .
$\mathcal{W}_r, \mathcal{W}_i, \mathcal{W}_\varphi$	Measurement noises of bistatic range, bistatic velocity, and azimuth, respectively.
T_S	Sample interval.
$\sigma_r, \sigma_i, \sigma_\varphi$	Standard deviations of bistatic range, bistatic velocity, and azimuth, respectively.
R_k	Covariance matrix of the measurement noise.
$r_k, \dot{r}_k, \varphi_k$	Bistatic range, bistatic velocity, and azimuth at time, t_k respectively.
r, φ, L	Bistatic range, target azimuth, and baseline distance, respectively.

$\mathbf{t}, \mathbf{t}_k$	Target position vector and target position vector at time t_k .
$\mathbf{P}_{\text{dt}}$	Covariance of localization error.
$\mathbf{z}_{1:K}, \mathbf{z}_{1:K}^N$	Measurement sample sequence and normalized measurement sequence.
$\mathbf{z}, \hat{\mathbf{t}}(\cdot)$	Input measurement and target position function estimated by neural network.
η, N	Smoothing factor and sample size.
$\mathbf{z}_k^{m_N}, \mathbf{z}_k^{n_N}$	Training sample and testing sample acquired by cross-validation at time t_k .
$\hat{\mathbf{t}}_{k,i}$	The estimated position vector of the i -th Monte Carlo simulation at time t_k .
M_C	Number of Monte Carlo simulations
η^{m_i}	Smoothing factor corresponding to training sample m_i for the i -th iteration.
$\hat{\mathbf{t}}_k$	Position measurement estimated by localization model.
$\mathbf{F}$	transition matrix
$\boldsymbol{\mu}_k$	System noise at time t_k .
$\hat{\mathbf{x}}_{k,i}$	The estimated state vector of the i -th Monte Carlo simulation at time t_k .

I. INTRODUCTION

PASSIVE radar, also known as passive coherent location, is a variant of bistatic radar. The passive radar system exploits noncooperative IOs to detect and track the targets of interest. Since there is no need for the deployment and operation of dedicated transmitters, the passive radar system has the benefits of a covert operation, low-cost nature, and spectrum saving. All these features have attracted increasing interest worldwide [1], [2], [3]. It has been widely applied in aviation surveillance, stealth aircraft detection, and long-range early warning of maritime targets [4], [5], [6], [7]. The performance of such applications requires the target tracking accuracy to be as high as possible. Accurate information on location and trajectory helps enhance airspace management. Nevertheless, localization and tracking performance in a passive bistatic radar system is limited. Hence, how to improve the target tracking performance remains an open and very challenging problem [8], [9].

The target localization and tracking, in essence, estimate the target position (localization problem) and the corresponding track (tracking problem) using the target measurements acquired from radar. Thus, the tracking problem can be described as the solution to a set of localization problems at successive time intervals. The existing localization technology in passive bistatic radar is mainly the GL method [10]. However, due to measurement nonlinearity, receiver placement error, and low azimuth accuracy, the localization performance with the GL approach is unstable and decreased. Many localization algorithms have been proposed in recent decades to address the above-mentioned problems. In [11], [12], and [13], the multistatic passive radar localization technology is used to improve the detection accuracy. It achieves higher precision by fusing the information from multiple bistatic pairs to eliminate the influence of azimuth accuracy. In this article, we only consider target tracking based on a bistatic radar

system, which is more economical. In [14], the receiver placement is optimized to improve the localization performance. In [15], [16], and [17], the measurement-conversion methods are used to obtain the pseudolinear measurements. It converts nonlinear measurements into Cartesian coordinates. The pseudolinearization method mainly aims at passive detection systems with only angle measurement and requires high accuracy of angle measurement. In addition, the approximation of nonlinear functions or the probability-density-function-based approaches are developed in [18], [19], and [20] to deal with the nonlinearity of the measurement. These methods cannot solve the problem well owing to the sensitivity of the initial value and the high-accuracy requirements of noise statistics. Therefore, localization approaches capable of filtering out the measurement noises are desired to tackle the nonlinearity problem between the target and measurements under a dynamic environment.

Recently, there has been a trend of using deep learning for localization and tracking [21], [22], [23], [24], [25], which can deal with noisy measurements and can learn and generalize very quickly. Especially when the mapping relationship between the input and the output of the system is unclear or affected by noise, they are widely adopted. Unlike the traditional localization and tracking methods, the localization approach based on an NN does not need prior knowledge of noise distribution. This leads to the significant advantages of radar systems detecting targets with NN techniques. NNs are mainly used in the node localization field [26], [27], [28], [30]. In [26], convolutional NNs (CNN) are used to recognize the 3-D radio images constructed by received signal strength indicator (RSSI) fingerprints. Then a sensor node location can be acquired. But these deep NN-based tracking approaches are designed for images. They are not suitable for radar tracking with measurements. Deep long short-term memory network (LSTM) is developed in [27] to track different types of targets. The deep LSTM requires relatively more artificial adjustment parameters. In [28], the TDOA localization method based on RBF is proposed for accurate node localization. It needs more sample data, and the RBF network has higher localization complexity and lower efficiency. In [29], the localization model based on multilayer NN is used to reduce the localization error. It is difficult to apply to large-scale networks due to high localization complexity and many optimization parameters. In [30], the GRNN replaces this traditional RSSI-based approach to estimate the first location of the target for wireless sensor networks. However, this model is subject to the smoothing parameter factors' selection of GRNN, which has problems such as low learning efficiency and significant subjective influence of human beings. Some related works from the areas are shown in Table I.

As seen from the above analysis, we found that GRNN has the strong nonlinear mapping ability of the RBF network and has good fault tolerance and robustness. GRNN has a fast convergence speed and needs fewer sample data compared with the above-mentioned NNs. To solve the nonlinearity problem of the target and the measurement, a novel target tracking algorithm for passive bistatic radar tracking is proposed. The algorithm treats localization as a regression

TABLE I
COMPARISON OF DIFFERENT MODELS FOR TARGET TRACKING

	Main distinctive characteristics	
Model-based Target Tracking	Localization (GL) with tracking (KF) Directly tracking: EKF and UKF based tracking method	
Data-driven Target Tracking	Sequence Tracking	A modified RNN-based structure is shown to be able to output both variance and mean of the estimation. [22] RNN/LSTM-based deep network structure is employed to perform prediction and filtering. [27]
	Localization Tracking	A method based SVM and twin SVM is used to detect region of the desired event (source localization). [25] CNN is used to recognize the 3D radio images constructed by RSSI fingerprints, and obtain determined location region of sensor node (location recognition problem). [26] The distance information, which is calculated by TDOA data, is input of RBF network. Position of sensor node is target output (node localization). [28] The localization problem is transformed into that of classification. Multi-layer NN improve localization error by designing a boosting method (door localization). [29] Input to the GRNN architecture is the RSSI measurements, the output of the network is target position. It can be trained quickly using very few training samples.[23], [30]

problem and uses GRNN to model the dependency between the target measurement and the position. The inputs to the GRNN model are the target measurements obtained from the radar, and the outputs are the 2-D target positions. To eliminate the influence of the sample size and smoothing factor on the performance of the GRNN model, the selection of the input sample size and smoothing factor of the localization model is transformed into an optimization problem. The RMSE of the detection area location is taken as the objective function. We propose to apply a CCGA [31] to the GRNN localization model. In addition, the method applies the KF framework on the CCGA-GRNN localization results to correlate consecutive location estimations and form precise trajectories. The simulations and field experimental data of the passive radar demonstrate the performance of the proposed algorithm.

The meanings of the notation in the article are listed in the Nomenclature. The main contributions of the article are as follows.

1) A fast optimization-based target tracking model is developed using CCGA. The smoothing factor of the GRNN is commonly set by professional experience, which is subjective. Here, the influence of the sample size and smoothing factor on the GRNN model performance is considered. The CCGA method is introduced to solve the problem. The proposed algorithm can implicitly extract features from the input with less sample size and optimum smoothing factor, achieving higher tracking accuracy.

2) An accurate regression model is efficiently established by exploiting GRNN/CCGA-GRNN for target feature extraction and position estimation. As mentioned above, GRNN/CCGA-GRNN outperforms the commonly used regression methods. In addition, we apply KF on the GRNN/CCGA-GRNN localization results to form precise trajectories of targets.

The rest of this article is organized as follows. The localization accuracy analysis and measurement model are presented in Section II. Section III describes the proposed tracking

algorithm, which mainly includes measurements preprocessing, localization with GRNN regression, CCGA optimization, and tracking with the Kalman structure. Section IV demonstrates the performance of the proposed algorithm via numerical simulations and experimental data processing. Finally, conclusions are drawn in Section V.

II. PROBLEM FORMULATION

Consider a 2-D scenario in which one transmit and one receive antennas exist. The receiver is placed at a known position. The objective of this article is to accurately determine the target position using the collected information from radar. The information includes bistatic range, bistatic velocity, and the azimuth of a target. To this end, the target localization accuracy for bistatic radar is first analyzed, and then the target measurement function is introduced in this section.

A. Accuracy Analysis

A passive radar system with a single receiver Rx and single transmitter Tx is taken as an example here. Their positions are known, denoted by $\mathbf{s}_r = [x_r, y_r]^T$ and $\mathbf{s}_t = [x_t, y_t]^T$, respectively, where “T” is the transpose operation. The target measurement at time t_k is defined as

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_k) + \boldsymbol{\omega}_k = [r_k, \dot{r}_k, \varphi_k]^T + \boldsymbol{\omega}_k \quad (1)$$

where $\mathbf{x}_k$ is the target state at time t_k , $\mathbf{z}_k$ is the target measurement vector detected by the radar at time t_k , $\mathbf{h}$ is the measurement function associated with the bistatic pair Tx-Rx, and $\boldsymbol{\omega}_k$ is the measurement noise. It is defined as

$$\begin{aligned} \boldsymbol{\omega}_k &= [w_r, w_{\dot{r}}, w_{\varphi}]^T, \quad w_r \sim \mathcal{N}(0, \sigma_r^2) \\ w_{\dot{r}} &\sim \mathcal{N}(0, \sigma_{\dot{r}}^2), \quad w_{\varphi} \sim \mathcal{N}(0, \sigma_{\varphi}^2) \end{aligned} \quad (2)$$

where $\mathcal{N}$ denotes the normal distribution, and the bistatic range noise w_r follows a normal distribution with zero-mean and variance σ_r^2 . Likewise, bistatic velocity noise $w_{\dot{r}}$ and

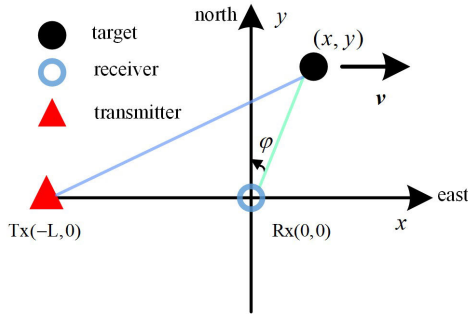


Fig. 1. Geometric diagram of passive radar system with one transmitter and one receiver in 2-D space.

azimuth noise w_φ follow a normal distribution with zero-mean and variance σ_r^2 and σ_φ^2 , respectively. The covariance matrix of ω_k conforms to

$$E\{\omega_k \omega_k^T\} = \mathbf{R}_k \delta_{kj} \quad (3)$$

where $E\{\cdot\}$ denotes the statistical expectation, δ_{kj} is the Kronecker function, and $\mathbf{R}_k$ is the symmetric positive definite covariance matrix of the measurement noise ω_k at time t_k , i.e.,

$$\mathbf{R}_k = \begin{bmatrix} \sigma_r^2 & 0 & 0 \\ 0 & \sigma_r^2 & 0 \\ 0 & 0 & \sigma_\varphi^2 \end{bmatrix}. \quad (4)$$

The geometry of the passive radar system is depicted in Fig. 1. The coordinate system is established with the receiver at the origin, x-axis to the east, and y-axis toward the north. The coordinate of the transmitter Tx is $s_t = [x_t, y_t]^T = [-L, 0]^T$, where L denotes the baseline distance. The main goal of the radar system is to obtain the accurate location of the target. The target position (x, y) can be deduced by the GL method

$$\begin{cases} x = r_1 \sin \varphi = \frac{(r^2 - L^2) \sin \varphi}{2(r + L \sin \varphi)} \\ y = r_1 \cos \varphi = \frac{(r^2 - L^2) \cos \varphi}{2(r + L \sin \varphi)} \end{cases} \quad (5)$$

where $r_1 = \sqrt{x^2 + y^2}$ is the distance between the target and the receiver. r is the bistatic range, and φ is the azimuth.

The position estimation accuracy is very important for target tracking. Denote the position of the target by $\mathbf{t} = [x, y]^T$, and assume that $\mathbf{m} = [r, \varphi]^T$. According to (3), measurement noises at different times are assumed to be statistically independent. We have

$$d\mathbf{t} = \begin{bmatrix} dx \\ dy \end{bmatrix} = \begin{bmatrix} \partial x / \partial r & \partial x / \partial \varphi \\ \partial y / \partial r & \partial y / \partial \varphi \end{bmatrix} \begin{bmatrix} \sigma_r \\ \sigma_\varphi \end{bmatrix} = \mathbf{C} \cdot d\mathbf{m}. \quad (6)$$

The covariance of location error is given by

$$\mathbf{P}_{dt} = E\{d\mathbf{t} d\mathbf{t}^T\} = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix} = \mathbf{C} E\{d\mathbf{m} d\mathbf{m}^T\} \mathbf{C}^T \quad (7)$$

where

$$E\{d\mathbf{m} d\mathbf{m}^T\} = \begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\varphi^2 \end{bmatrix}. \quad (8)$$

It can be seen that the position accuracy is explicitly influenced by the bistatic time estimation error and azimuth

error, when the receiver placement is determined. To reduce the uncertain noise effect involved in target measurements on the localization performance, it is crucial to establish an accurate target localization model that matches the target's actual motion position. As mentioned in the previous section, the existing GL method differs in the selection of measurement information and the treatment of the highly nonlinear measurement relationship (see Section II-B). Nevertheless, other new methods based on accurate system models may not handle the nonlinearity well, which makes it difficult for these methods based on model-driven to adapt to application requirements. Hence, the accurate localization model with more measurement information instead of traditional methods is used in this article to estimate the target position.

B. Measurement Model

As shown in Fig. 1, assume there is a target moving with constant velocities (v_x, v_y) in the Cartesian coordinates. According to the assumption of constant velocity motion in the Cartesian coordinates, the target position in the x- and y-directions at time t_k can be expressed as

$$\begin{cases} x_k = x_0 + v_x t_k \\ y_k = y_0 + v_y t_k \end{cases} \quad (9)$$

where (x_0, y_0) indicates the initial position of the target. $t_k = (k - 1)T_s$ ($k = 1, 2, \dots, K$) denotes the slow time, K and T_s represent the number of sample and sample interval, respectively.

Then, the bistatic range r at time t_k can be represented as

$$r_k = \sqrt{(x_k)^2 + (y_k)^2} + \sqrt{(x_k + L)^2 + (y_k)^2}. \quad (10)$$

Define $a_k = \sqrt{(x_k)^2 + (y_k)^2}$, $b_k = \sqrt{(x_k + L)^2 + (y_k)^2}$. The bistatic velocity can be given by

$$\dot{r}_k = \dot{a}_k + \dot{b}_k. \quad (11)$$

Substituting (9) and (11) into (10) yields

$$r_k = \sqrt{a_0^2 + 2a_0\dot{a}_0 t_k + v^2 t_k^2} + \sqrt{b_0^2 + 2b_0\dot{b}_0 t_k + v^2 t_k^2} \quad (12)$$

where $v = \sqrt{v_x^2 + v_y^2}$ indicates the constant velocity of the target. The target azimuth can be expressed as

$$\varphi_k = \text{atan2}(x_k, y_k) \quad (13)$$

where $\text{atan2}(\cdot, \cdot)$ denotes the four-quadrant inverse tangent.

The target measurement function is

$$\mathbf{h}(\mathbf{x}_k) = [r_k, \dot{r}_k, \varphi_k]^T. \quad (14)$$

It can be seen that the time evolution of measurements of the target is a highly nonlinear function with the square root over the 2nd-order polynomial. Note that the existing model-driven-based localization methods cannot absorb measurement nonlinearity well. The nonlinear relationship between the target and the measurements in (1) will affect the position accuracy of the target for classical methods. Especially in the case of complex measurement noise, its localization performance will be degraded. This article aims to accurately determine the target position from these noisy measurements.

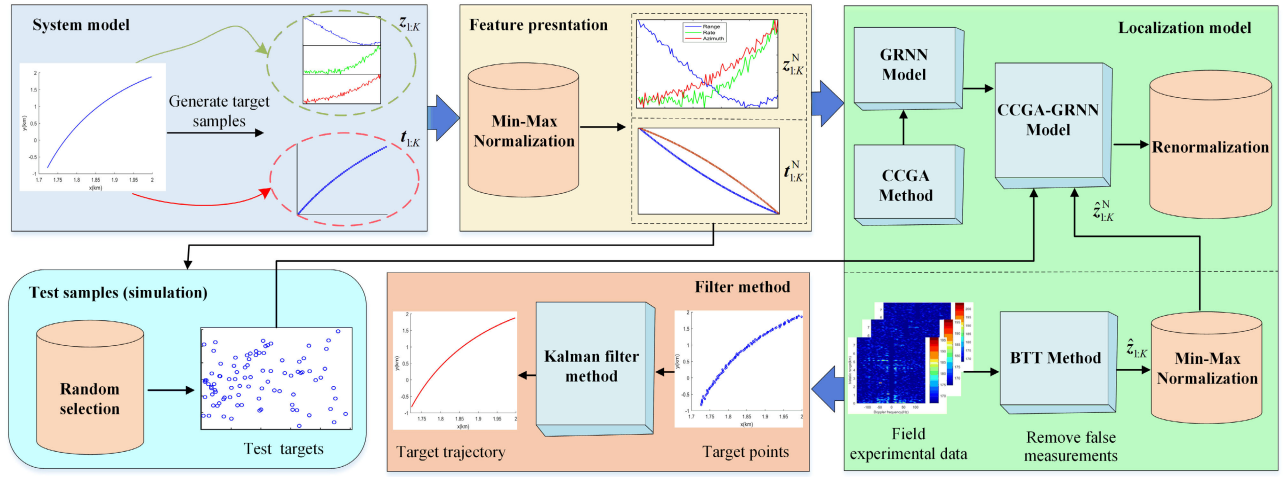


Fig. 2. Diagram of the proposed GRNN-based target tracking.

III. GRNN-BASED TARGET TRACKING

In this section, a novel GRNN-based tracking algorithm is developed. The overall architecture of the proposed algorithm is illustrated in Fig. 2. One can see that K target samples around the current detection area are first generated, and then the min-max normalization technique is performed to modify the samples to make them suitable for training. This can alleviate the problem that the data features of the bistatic velocities and azimuth are submerged by the bistatic range data. With the learned features, the CCGA-GRNN localization model is trained for target localization. When the original measurements are used to validate the trained localization model, the original measurement is first preprocessed by the BTT algorithm [32] to remove false measurements. The retained measurements can be used to realize the one-step-ahead estimation of the target position. The KF method recursively updates the position according to the current location estimation returned by CCGA-GRNN. The final smoothing trajectories can then be obtained.

A. Measurement Preprocessing

Raw target measurement data contain false measurement information in radar detection. We apply the BTT algorithm to remove false measurements. The min-max normalization method is adopted to ensure that measurement features have the same metric scale.

1) *Remove False Measurement Information*: Raw measurement data have false alarms. We apply the BTT algorithm to detect and remove the false target. BTT is the framework of the bistatic-coordinate-based processing method, which can recognize the sequential target measurements and make use of this sequential relationship to associate target measurements and remove false measurements. The remained measurements $\hat{z}_{1:K}$ after BTT processing are suitable for tracking (see Fig. 2).

2) *Normalization*: Each measurement z_k contains different measurement features: bistatic distance, bistatic velocity, and azimuth, which have different dimensions. When the measurements are fed into the localization network, the

features' information of the bistatic velocity and azimuth may be submerged by the bistatic range data because they are smaller than the bistatic range. This affects the localization network output. To eliminate the dimension influence between the input measurements, the min-max normalization technology is used to modify the input. Measurement $z_{1:K} = \{z_i : i = 1, 2, \dots, K\}$ is normalized to generate the input $z_{1:K}^N = [\tilde{r}_1^N, \dots, \tilde{r}_K^N, \tilde{r}_1^N, \dots, \tilde{r}_K^N, \tilde{\varphi}_1^N, \dots, \tilde{\varphi}_K^N]$, where

$$\tilde{r}_i^N = (2(\tilde{r}_i - \min(\tilde{r}_{1:K}^N)) / (\max(\tilde{r}_{1:K}^N) - \min(\tilde{r}_{1:K}^N))) - 1 \quad (15)$$

$$\tilde{r}_i^N = (2(\tilde{r}_i - \min(\tilde{r}_{1:K}^N)) / (\max(\tilde{r}_{1:K}^N) - \min(\tilde{r}_{1:K}^N))) - 1 \quad (16)$$

$$\tilde{\varphi}_i^N = (2(\tilde{\varphi}_i - \min(\tilde{\varphi}_{1:K}^N)) / (\max(\tilde{\varphi}_{1:K}^N) - \min(\tilde{\varphi}_{1:K}^N))) - 1 \quad (17)$$

where “max” and “min” are the maximum absolute value and the minimum absolute value in the elements of the input vector, respectively. Meanwhile, the target position $t_k = [x_k, y_k]^T$ is also processed by the same normalization. It scales the real-time trained data $(z_{1:K}, t_{1:K})$ to values $(z_{1:K}^N, t_{1:K}^N)$ between -1 and 1 , which will be used as the input for subsequent model training and tracking.

B. Localization With GRNN Regression

Recall the measurement model we have adopted in (14). We now discuss how to accurately convert the nonlinear measurements into the target position in the Cartesian coordinates. It can be solved as a regression problem by establishing the dependency between the target measurements and the target positions. GRNN has been proven to be very effective in implicit feature extraction and has gained vast applications in machine learning. We model the dependency between the target position and the measurement with GRNN and formulate localization as a regression problem. The inputs to the GRNN model are the preprocessed measurements $z_{1:K}^N$, and the outputs are the target position t_k .

Localization through GRNN regression consists of the training stage and the localization stage. In the training stage,

a large number of track samples are simulated based on the CA model [33] in the detection area of interest. The GRNN model is then trained using the measurements of all the targets labeled with their positions. This is to determine the smoothing factor of the NN. In the localization stage, the retained measurements after preprocessing $\hat{z}_{1:K}^N$ are fed into the trained GRNN model, which outputs the 2-D target position.

Suppose that the training dataset is $D = \{(z_k, t_k) : k = 1, 2, \dots, K\}$. A common way of using D to calculate an approximation $\hat{t}(z)$ of a function $\phi : z \in \mathbb{R}^3 \rightarrow t \in \mathbb{R}^2$ is to minimize the L_2 risk $E[(t - \hat{t}(z))^2]$. This can be shown to satisfy the conditional mean regression $\hat{t}(z) = E[t|z]$, which expands to

$$\hat{t}(z) = \int t p(t|z) dt = \int t p(t, z) dt / \int p(t, z) dt \quad (18)$$

where $p(t|z)$ denotes the conditional output probability density.

The output $\hat{t}(z)$ is calculated using an approximation $\hat{p}(t, z)$ of the real but unknown joint density. The key idea is that this approximation is based on the observations contained within the training set and a nonparametric kernel-based estimate modeled with PW [34], [35]. According to PW, the joint density function $\hat{p}(t, z)$ can be expressed as

$$\hat{p}(t, z) = \frac{\sum_{k=1}^K \exp[-\frac{(z-z_k)^T(z-z_k)}{2\eta^2}] \exp[-\frac{(z-t_k)^2}{2\eta^2}]}{K(2\pi)^2\eta^4} \quad (19)$$

where η represents the bandwidth coefficient of the Gaussian window function. It also called be smoothing factor SPREAD. Then we substitute $\hat{p}(t, z)$ for $p(t, z)$ into (18) and swap the order of integration and summation. The position output $\hat{t}(z)$ can be defined as

$$\hat{t}(z; \eta, D) = \frac{\sum_{k=1}^K t_i \exp(-\frac{(z-z_k)^T(z-z_k)}{2\eta^2})}{\sum_{k=1}^K \exp(-\frac{(z-z_k)^T(z-z_k)}{2\eta^2})}. \quad (20)$$

It can be noted that the position estimation $\hat{t}(z)$ is the weighted average of all the measurement samples z_k , where the weight for each measurement is the exponential of the squared Euclidian distance between samples z and z_k . Hence, the regression prediction of $\hat{t}$ is yielded by feeding the input data z to the estimation network. Then, the accurate position of the target can be obtained.

As shown in Fig. 3, the estimation network of the target position is developed of the input layer, pattern layer, summation layer, and output layer. The number of neurons in the pattern layer is usually equal to the training sample number. The summation layer has two types of neurons such as $S_{k,D}$ and $S_{k,i}$ $i = 1, 2$, where

$$S_{k,D} = \sum_{k=1}^K \exp\left(-\frac{(z-z_k)^T(z-z_k)}{2\eta^2}\right). \quad (21)$$

and

$$S_{k,i} = \sum_{k=1}^K t_i \exp\left(-\frac{(z-z_k)^T(z-z_k)}{2\eta^2}\right) \quad (22)$$

denote the transition function of the summation layer at frame k . The number of neurons in summation is $i + 1$.

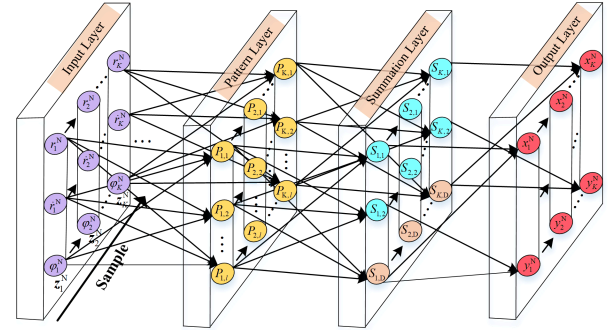


Fig. 3. Block diagram of the GRNN architecture for the proposed target tracking. In this structure, the input data flow along with the arrows through all the layers and finally become the estimation of the target position t_k .

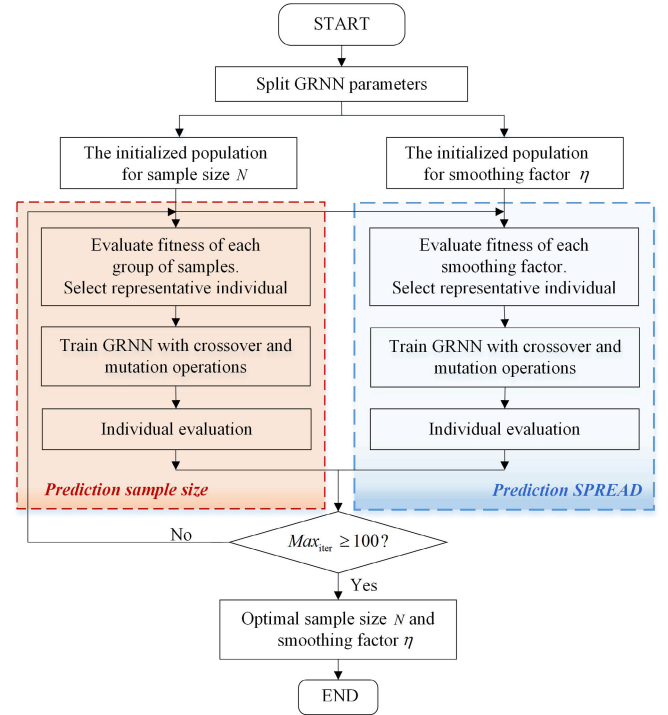


Fig. 4. Flowchart of the CCGA-GRNN method.

C. CCGA Optimization

The selection of sample size and smoothing factor SPREAD is closely related to the performance of the GRNN model. They are generally tuned by empiricism, which has a large impact on estimation accuracy. In this article, CCGA with strong global search ability is introduced into GRNN to optimize the sample size and SPREAD. It can improve the estimation performance and training efficiency of GRNN. The specific steps of the CCGA algorithm, as shown in Fig. 4, are as follows.

Step 1: The separate populations of sample size and smoothing factor are initialized, respectively. $N_{\min}$ and $N_{\max}$ denote the minimum and maximum values of the sample size, respectively. $\eta_{\min}$ and $\eta_{\max}$ indicate the minimum and maximum values of the smoothing factor, respectively. Then the initial population P_1 and P_2 are constructed using variables N and η , respectively, i.e., $P_1(t_k) = [N_{t_k}]$, $P_2(t_k) = [\eta_{t_k}]$.

Step 2: The representative individuals are selected for each subpopulation to form a cooperative group. Their respective GRNN can be obtained. The fitness function optimized by CCGA is designed to evaluate the superiority and inferiority of individuals. Since the RMSE can better reflect the deviation between the estimated value and the real value, the RMSE of the estimated value and the real value is used as the fitness function in this article. To improve the estimation capability of the GRNN, the samples are divided into m and n groups by adopting the cross-validation method, where m is the fitting sample corresponding to the population P_1 , and n is the test sample. The function expression with N and η as independent variables is given by

$$\hat{t}(z_k^{mN}, \eta) = \frac{\sum_{k=1}^K t_i^{mN} \exp\left(-\frac{(z-z_k^{mN})^T(z-z_k^{mN})}{2\eta^2}\right)}{\sum_{k=1}^K \exp\left(-\frac{(z-z_k^{mN})^T(z-z_k^{mN})}{2\eta^2}\right)}. \quad (23)$$

The fitness function can be expressed as

$$E_k(z_k^{nN}, \eta) = \sqrt{\frac{1}{M_C} \sum_{i=1}^{M_C} (t_k - \hat{t}_{k,i}(z_k^{nN}, \eta))^2} + \varepsilon \quad (24)$$

where t_k denotes the real value at time t_k , and $\hat{t}_{k,i}$ represents the estimated value of the i -th MC simulation at the same time. M_C is the number of MC simulations.

Step 3: According to the initial population P_1 and P_2 , and the fitness function E_k , the fitness function value of each individual is calculated. Then the function values are ranked in ascending order to obtain the best individual in the initial population. It is the current optimal individual. When the best individual is obtained by cross-validation, the training sample is recorded as m_1 , and the corresponding smoothing factor is η^{m_1} .

Step 4: The GRNN model is optimized by the crossover and mutation to obtain the optimal individual. New subpopulations are generated after evolution.

Step 5: The maximum algebra method is adopted for the termination condition. The maximum iteration algebra is set as 100. If the termination condition is met, the evolution is stop. The corresponding small sample training set and the optimal smoothing factor ($m_{100}, \eta^{m_{100}}$) are output. Otherwise, $t_k = t_k + 1$, skip step 2 to continue the cycle.

Step 6: The best $\eta^{m_{100}}$ obtained is taken as the SPREAD of GRNN. The best sample size N corresponding to m_{100} is obtained by cross-validation. It is used as the training sample to reconstruct the GRNN model. Then the measurement data to be tested are input into the CCGAG model, and the best target position can be obtained.

D. Tracking With Kalman Structure

As aforementioned, the point localization may not form smoothing trajectories. For the problem, the target localization estimated by GL, GRNN, and CCGA-GRNN regression is regarded as the measurement. This article uses the KF method to test the GL, GRNN, and CCGA-GRNN-based tracking performance. Assume $\mathbf{x}_k$ represents the target state at time t_k . $\hat{\mathbf{t}}_k$ denotes the measurement (i.e., the estimated location

by GL, GRNN, and CCGA-GRNN regression, respectively) at time t_k . KF is to recursively estimate the posterior distribution $p(\mathbf{x}_k | \hat{\mathbf{t}}_{1:k})$ the target state $\mathbf{x}_k$ given the measurements $(\hat{\mathbf{t}}_1, \hat{\mathbf{t}}_2, \dots, \hat{\mathbf{t}}_k)$.

The state equation here can be expressed as

$$\mathbf{x}_k = \mathbf{F}(\mathbf{x}_{k-1}) + \boldsymbol{\mu}_k \quad (25)$$

where $\boldsymbol{\mu}_k$ represents the system noise assumed to be Gaussian. $\mathbf{F}$ is the state transition matrix as given below. The location $\hat{\mathbf{t}}_k = [x^z y^z]^T$ estimated by the aforementioned three algorithms is used as the measurement. According to the definition of the CA shape, the state transition matrix $\mathbf{F}$ in (25) is $\mathbf{F} = \text{blkdiag}(\mathbf{F}_1, \mathbf{F}_1)$, where

$$\mathbf{F}_1 = \begin{bmatrix} 1 & T_S & T_S^2/2 \\ 0 & 1 & T_S \\ 0 & 0 & 1 \end{bmatrix}. \quad (26)$$

The KF method contains the predicting and updating stages. In the predicting stage, the estimation from the previous time $t_k - 1$ is used to produce an estimation of the current time t_k . Whereas in the updating stage, the location measurements from the current time t_k are exploited to refine the estimation of the predicting step to improve it.

IV. SIMULATIONS AND FIELD EXPERIMENT RESULTS

In this section, the simulations are first conducted to evaluate the performance of the proposed algorithm. Real data are then used to test the practical performance of the proposed approach.

To measure the accuracy and robustness of the proposed algorithm, the RMSE and MRE are used for tracking performance evaluation in this article. They are calculated as follows:

$$\text{MRE}_k = \frac{1}{M_C} \sum_{i=1}^{M_C} \left(\frac{1}{K} \sum_{k=1}^K \frac{\|\mathbf{x}_k - \hat{\mathbf{x}}_{k,i}\|}{\|\mathbf{x}_k\|} \right) \quad (27)$$

$$\text{RMSE}_k = \sqrt{\frac{1}{M_C} \sum_{i=1}^{M_C} \|\mathbf{x}_k - \hat{\mathbf{x}}_{k,i}\|^2} \quad (28)$$

where MRE_k and RMSE_k are MRE and RMSE at time t_k , respectively. $\mathbf{x}_k$ represents the real value at time t_k . $\hat{\mathbf{x}}_{k,i}$ is the estimated value of the i -th MC simulation at the same time. The MRE can be used to evaluate the fluctuation degree of estimation error and to test the robustness and stability of the proposed model. RMSE is an absolute measure of fitting for the proposed tracking algorithm. The lower the results of the statistical indicators, the better the algorithm.

A. Simulation Setup

In this article, the target tracking database is established by simulating the system characteristics of passive bistatic radar. The simulation parameters are shown in Table II. The samples in the database contain the measurements and the trajectories' truth, which constitute the input-output pairs for the localization network. In all, 4000 tracks are simulated in the experiments, which are randomly and equally grouped into four batches, i.e., each batch contains 1000 track segments. About 1000 trajectories' truths selected randomly

TABLE II
SIMULATION PARAMETERS

Symbol	Parameter	Range
t_0	Initial target position	35 ~ 2200 (m)
v	Target speed	0 ~ 30 (m/s)
a	Target acceleration	0.01 ~ 5 (m/s ²)
w_r	Bistatic range noise	5 ~ 30 (m)
w_f	Bistatic velocity noise	0.5 ~ 2 (m/s)
w_ϕ	Azimuth noise	0.1° ~ 3°

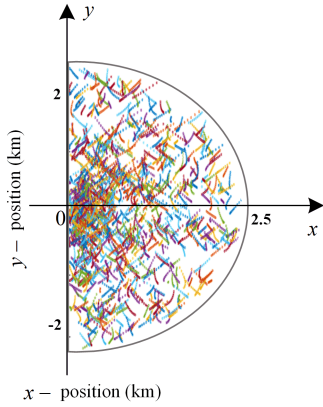


Fig. 5. Truth of target tracks. All truths of 1000 track segments are depicted in the same figure. (The length of each track segment is 10 s.)

TABLE III
LOCALIZATION ACCURACY WITH DIFFERENT METHODS

Method	GL	GRNN	CCGA-GRNN	ELM
RMSE (m)	88.31	74.94	42.66	76.98
MRE	0.068	0.062	0.021	0.066

are visualized in Fig. 5. As shown in this figure, the target trajectory segments in the target localization database cover all possible UAV detection fields. For CCGA optimization, the crossover probability is set as $p_c = 0.7$. The mutation probability is set to $p_m = 0.05$. The optimal sample size N and smoothing factor η are obtained by 10 000 sample data and 100 iterations of training. Thus, a stable CCGA-GRNN output can be obtained.

B. Localization Performance

In this section, we will validate the key components of the proposed tracking algorithm: GRNN-based localization and CCGA-GRNN-based localization. We also use the GL method and the ELM method in [36] for comparison. The comparisons are made in 180 randomly selected targets. In our numerical results, the system performance is measured in terms of the RMSE and MRE over 100 simulation runs.

1) *Compared With GL*: GL is a commonly used localization method. We replace CCGA-GRNN regression with the GL method and apply the GL method to locate the target using the same testing datasets. The localization results are illustrated in Fig. 6 and Table III. The GL method achieves an accuracy

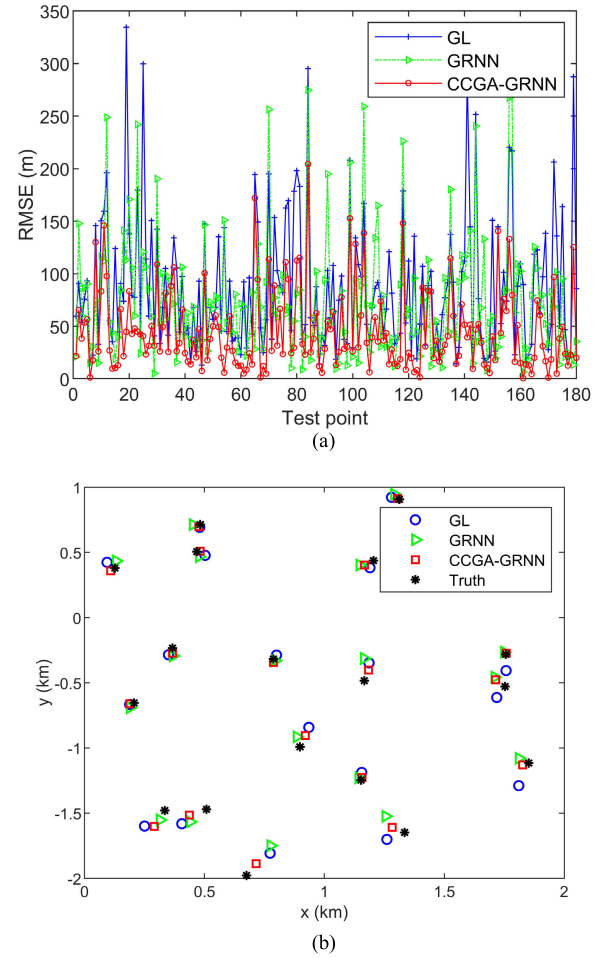


Fig. 6. Localization results with different algorithms. (a) Position RMSE comparison of different methods. (b) Results of localization by different methods (part test targets).

of 88.31 m, which is far less accurate than the CCGA-GRNN algorithm. Since GL is a geometry-based localization method, it is unable to deal with nonlinearity very well. Hence, compared with the GL method, CCGA-GRNN is more appropriate for passive bistatic radar localization.

2) *Compared With GRNN*: GRNN is a classic statistical tool for regression. For the CCGA-GRNN algorithm, the optimal smoothing factor is $\eta = 0.343$ by the iterations of training. The smoothing factor of GRNN is $\eta = 1$ by default. We apply GRNN to locate the target using the same training and testing datasets. The localization results are illustrated in Fig. 6 and Table III. GRNN achieves an accuracy of 74.94 m, worse than GRNN with CCGA. The proper smoothing factor η is important to the accuracy of GRNN in radar detection. In view of the aid of CCGA, the proposed algorithm outperforms GRNN.

3) *Compared With State-of-the-Art ELM*: ELM is a single hidden layer feedforward NN based on the generalized inverse matrix theory. We reimplement the method of Huang et al., which achieves an accuracy of 76.98 m. CCGA-GRNN outperforms it by a large margin, as shown in Table III. In addition, the localization results for different training sample sizes are shown in Fig. 7. The RMSEs of the two methods decline along

TABLE IV
SETTING OF THREE TRAJECTORIES

Measurement noise	Number	Prior target state x_0
Case 1: $\Delta r = 5 \text{ m}$, $\Delta \dot{r} = 1 \text{ m/s}$, $\Delta \varphi = 0.5^\circ$	1	$[2272.6\text{m}, 20.6\text{m/s}, 0.3\text{m/s}^2, 795.7\text{m}, -6.8\text{m/s}, -0.15\text{m/s}^2]$
Case 2: $\Delta r = 20 \text{ m}$, $\Delta \dot{r} = 2 \text{ m/s}$, $\Delta \varphi = 1^\circ$	2	$[1536.4\text{m}, 9.59\text{m/s}, 0.62\text{m/s}^2, 778.6\text{m}, 14.2\text{m/s}, -1.3\text{m/s}^2]$
Case 3: $\Delta r = 30 \text{ m}$, $\Delta \dot{r} = 2 \text{ m/s}$, $\Delta \varphi = 3^\circ$	3	$[1997.9\text{m}, 3.77\text{m/s}, -0.02\text{m/s}^2, 1878.1\text{m}, 16.4\text{m/s}, 0.22\text{m/s}^2]$

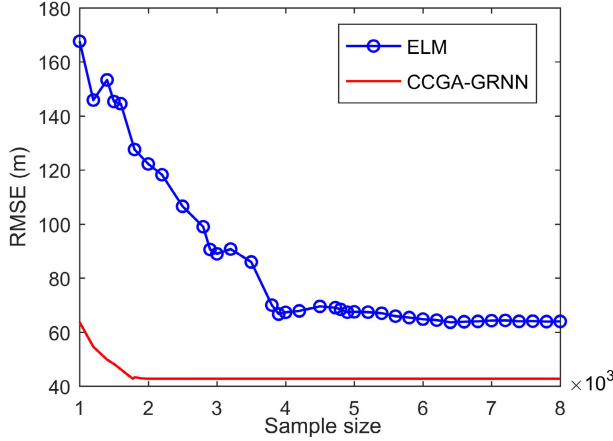


Fig. 7. Three test targets in the Cartesian coordinates.

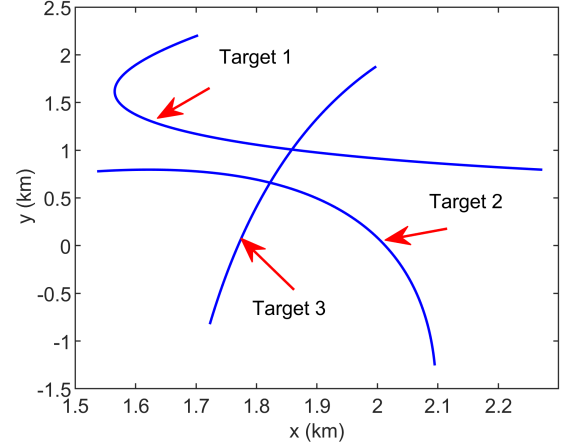


Fig. 8. Three test targets in the Cartesian coordinates (simulation data).

with the increase in training sample size. Compared with ELM, the proposed algorithm has higher localization accuracy under a small number of training samples, which can save the cost. This shows the superiority of the CCGA-GRNN model.

C. Tracking Performance

In this section, we measure the tracking accuracy of the proposed algorithm against the other ones. In view of the performance of the ELM method being worse than that of the GRNN approach, other three tracking methods are used as competitors here: GL-based estimation, unscented Kalman filter (UKF)-based estimation, and GRNN-based estimation. The performance comparison of the methods is conducted in three cases with different measurement conditions. Three target trajectories are simulated for each measurement condition (see Fig. 8). Specifically, each trajectory lasts for 100 s, whose parameters are defined in Table IV. The tracking performance is measured in terms of RMSE and MRE over 1000 simulation runs. For all the tracking techniques, the corresponding parameters have been set to the values maximizing the performance.

The comparison results of target position estimates of case 1 are displayed in Fig. 9. The proposed algorithm uses the localization model established by CCGA-GRNN to reduce the estimation error sufficiently. The proposed algorithm exhibits performance better than the other three methods. This is because the localization model used in the proposed tracking algorithm copes with the nonlinear measurements more accurately than the other methods.

In case 2, the standard derivations of measurement noises are increased. Fig. 10 shows similar results to case 1. It can be seen that the GRNN-based filter effectively reduces the

estimation error. Whereas the target position errors estimating from the GL method and UKF become pronounced, resulting in an obvious negative impact on the overall filtering accuracy after propagated in filtering recursions. On the contrary, the proposed algorithm has a stronger capability in dealing with nonlinearity and has higher estimation accuracy.

Fig. 11 shows the comparison results of position RMSEs for three trajectories. As the standard derivations of measurement noises increase from case 1 to case 3, the performance gap between the proposed method and the GL-based estimation and UKF-based estimation becomes more obvious. This means the proposed algorithm can better absorb the strong nonlinear between the target state and the measurements and uncertain measurement noise compared with the rest of the three algorithms. These results validate the effectiveness of the proposed method in a harsh situation with large measurement noises.

In addition, we investigate the means RMSE and MRE of the position tracking with all 1000 MC run for each trajectory. The results are displayed in Tables V and VI. The value Ga in bracket in Table V is the improvement in RMSE with the proposed algorithm compared with the other method. It is defined as

$$Ga = (\text{RMSE}_{\text{pr}} - \text{RMSE}_i) / \text{RMSE}_i, \quad (i = 1, 2, 3) \quad (29)$$

where RMSE_{pr} is the RMSE of the proposed algorithm. RMSE_i is the RMSE of the method i . $i = 1, 2, 3$ are the GL, UKF, and GRNN methods, respectively.

In the tables, the tracking RMSEs of the three algorithms are compared under different measurement noise conditions in Table IV. The smallest tracking RMSE and MRE are colored black in bold in each row, which shows that the corresponding

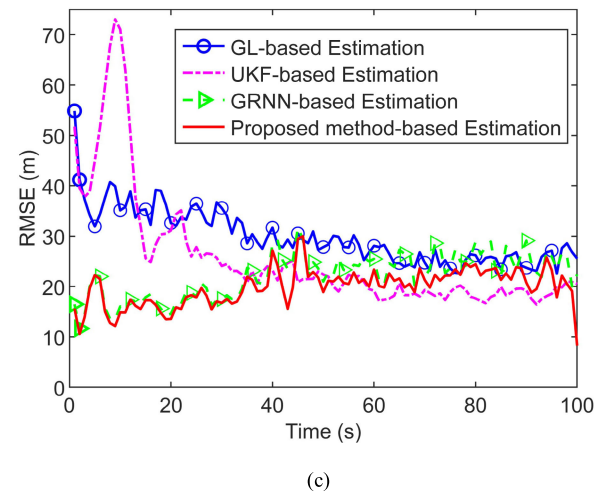
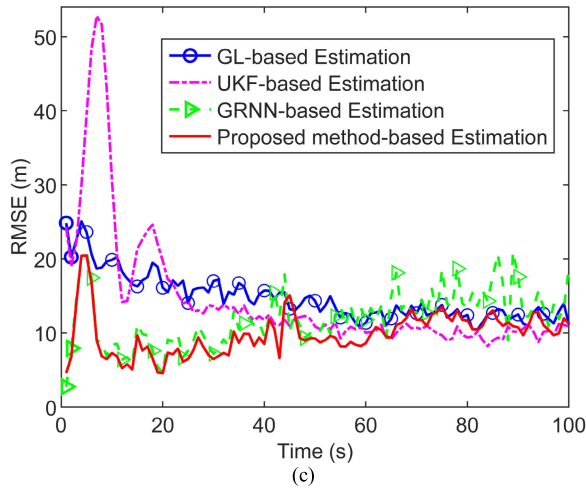
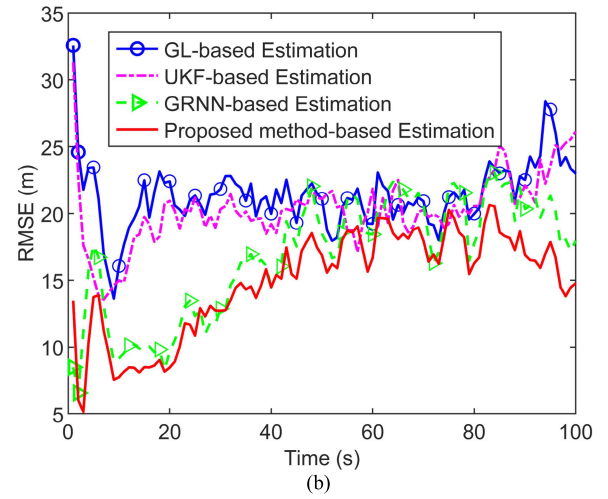
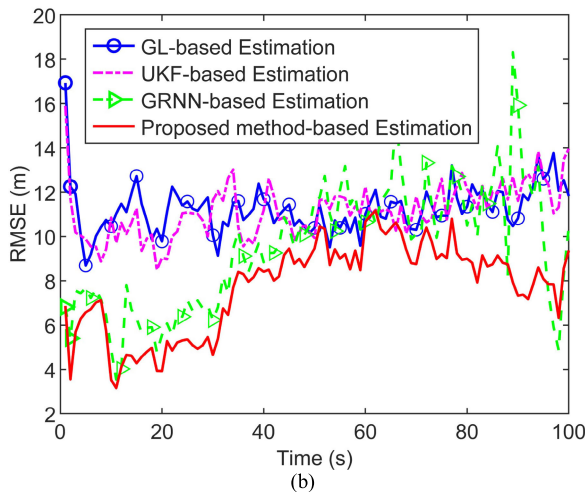
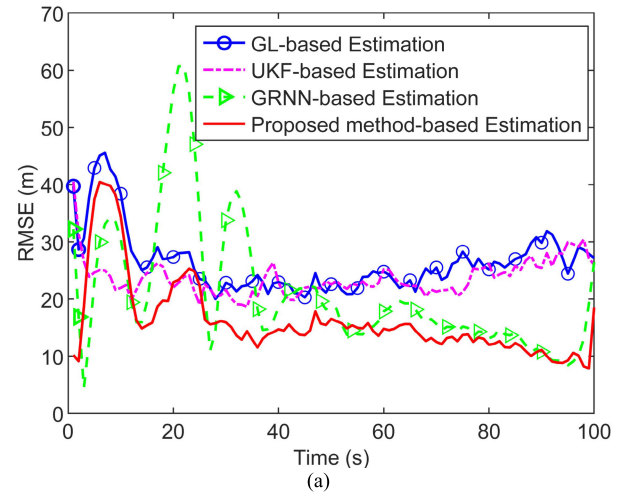
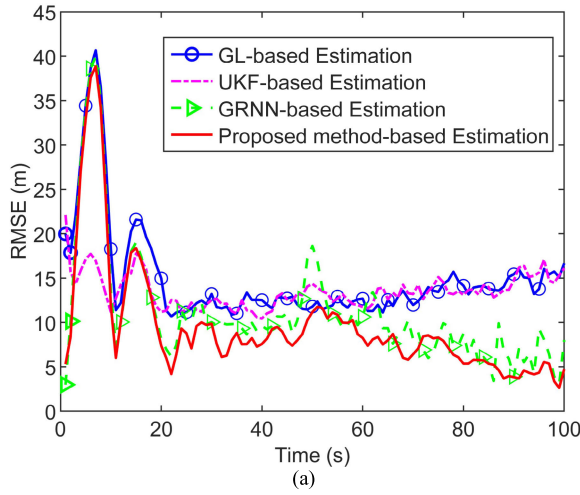


Fig. 9. Performance comparison of the three methods for three trajectories in case 1 ($\Delta r = 5\text{m}$, $\Delta \dot{r} = 1\text{m/s}$, and $\Delta \varphi = 0.5^\circ$). (a) RMSE plots of target 1. (b) RMSE plots of target 2. (c) RMSE plots of target 3.

Fig. 10. Performance comparison of the three methods for three trajectories in case 2 ($\Delta r = 20\text{m}$, $\Delta \dot{r} = 2\text{m/s}$, and $\Delta \varphi = 1^\circ$). (a) RMSE plots of target 1. (b) RMSE plots of target 2. (c) RMSE plots of target 3.

algorithm performs best. Obviously, all the tracking RMSEs and MREs of the proposed algorithm are black bold. The proposed algorithm provides the smallest tracking RMSEs and MREs. As we further observe in Table V, the RMSE of the proposed algorithm has a significant improvement compared

with the GL-based estimation. From case 1 to case 3, the proposed algorithm has 36.1%–53.82% improvement on RMSE. In summary, the proposed algorithm outperforms the other three approaches for target tracking.

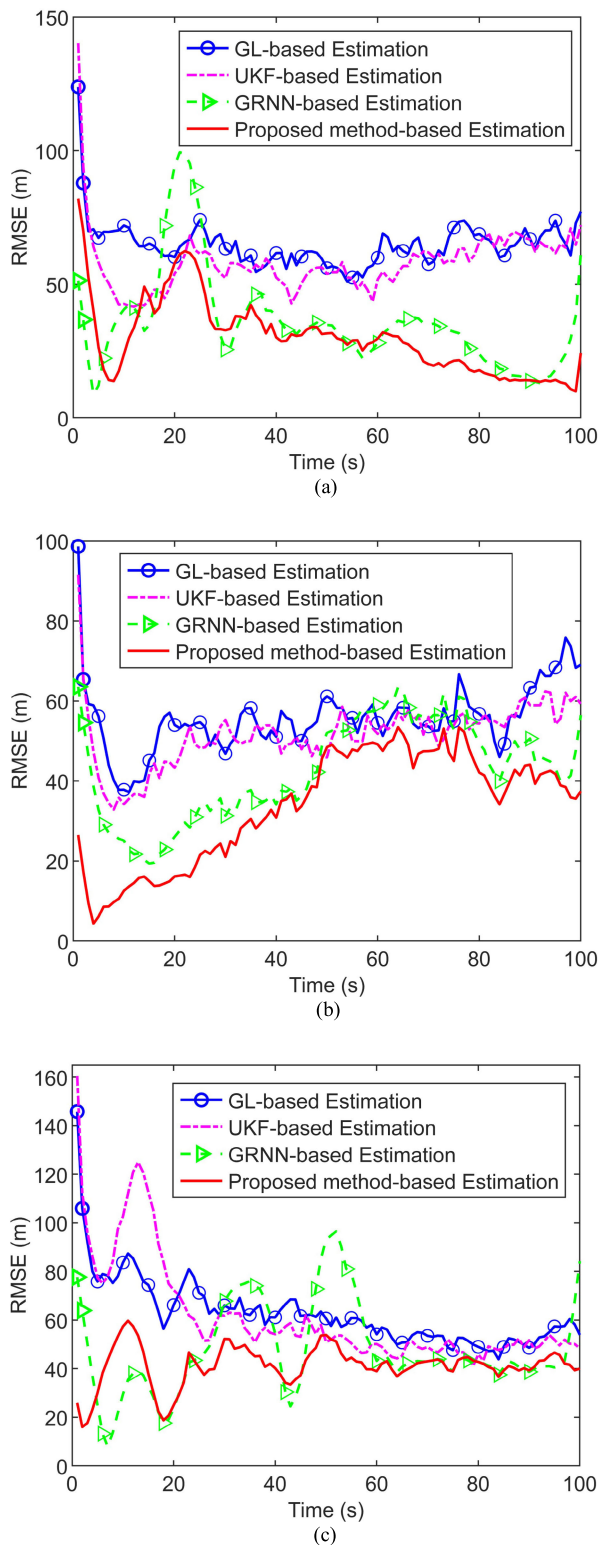


Fig. 11. Performance comparison of the three methods for three trajectories in case 3 ($\Delta r = 30\text{m}$, $\Delta \dot{r} = 2\text{m/s}$, and $\Delta \varphi = 3^\circ$). (a) RMSE plots of target 1. (b) RMSE plots of target 2. (c) RMSE plots of target 3.

D. Field Experimental Results

In this section, the practicability of the proposed method is demonstrated via a field experiment. The field experimental data, which are from an experimental passive bistatic radar system developed at Wuhan University in China, were

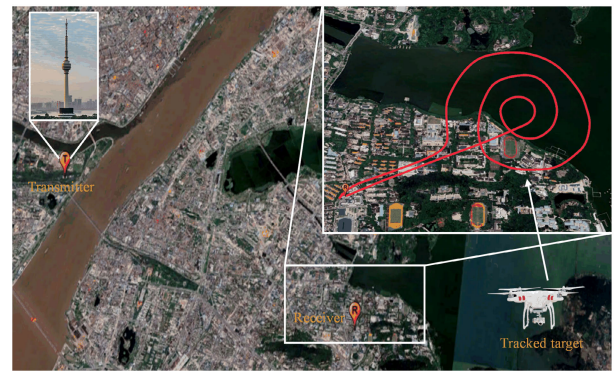


Fig. 12. Geometry of one transmitter, one receiver, and the area of interest.

acquired in July 2019. The selected duration was about 463 s. A quadrotor drone (named DJI phantom 4) with a diameter of about 0.5 m is used as the detection target. The data update period is 1 s. The experimental geometry is depicted in Fig. 12. There are one transmitter and one receiver in this experiment. The digital television (TV) signal is exploited as the illuminator of opportunity. The transmitter is Guishan Tower, and the receiver is located on the campus of Wuhan University. The transmitter Guishan Tower transmits digital TV broadcasting signal with the central working frequency of 754 MHz. The bandwidth is about 8MHz. The position information of the DJI drone recorded by the GPS is used as the reference to evaluate the tracking results.

Fig. 13 shows the target tracking results of this experiment. All the tracks obtained by the three tracking algorithms are quite continuous. As shown in Fig. 13(a), the tracking results achieve good agreement with the GPS from a 2-D view. The trajectory tracked by the proposed algorithm is closer to the GPS compared with the rest of the two tracking methods. This validates the feasibility of the proposed algorithm in practice. The position error is plotted in Fig. 13(b). The proposed method has a smaller position error, which means better tracking performance. This is because these location measurements converted by the proposed algorithm have higher accuracy. The trajectory in the Cartesian coordinates achieves superior performance using these rich performance converted measurements to update the trajectory.

The target position information recorded in the GPS module is used to calculate the RMSE of the trajectory. The results are listed in Table VII. The horizontal position accuracy and MRE of the proposed algorithm are 19.44 m and 0.023, respectively, while those of the GL-based estimation method are 37.23 m and 0.043, respectively, and those of UKF are 35.71 m and 0.038, respectively. The performance of the proposed algorithm is better than that of the GL and UKF methods. It validates that the proposed method performs well in field radar detection.

E. Computational Cost

In this section, the number of floating-point operations (FLOPs) is adopted as a measure of the model complexity with the proposed algorithm. Denote the dimension of the

TABLE V

MEANS OF THE POSITION TRACKING RMSE FOR ALL TRAJECTORIES WITH GL-BASED ESTIMATION, GRNN-BASED ESTIMATION, UKF-BASED ESTIMATION, AND PROPOSED ALGORITHM, RESPECTIVELY

	Tracks	Propose method(m)	GL (m) (Ga%)	UKF (m) (Ga%)	GRNN (m) (Ga%)
Case1	Track-1	9.54	14.93 (↓ 36.10%)	13.55 (↓ 29.59%)	11.08 (↓ 13.90%)
	Track-2	7.72	11.24 (↓ 31.32%)	11.17 (↓ 30.89%)	9.50 (↓ 18.74%)
	Track-3	9.87	14.75 (↓ 33.08%)	14.59 (↓ 32.35%)	12.03 (↓ 17.96%)
Case2	Track-1	16.18	26.40 (↓ 38.71%)	23.67 (↓ 31.64%)	21.09 (↓ 23.28%)
	Track-2	14.89	21.38 (↓ 31.36%)	20.27 (↓ 26.54%)	16.83 (↓ 11.53%)
	Track-3	19.83	29.77 (↓ 33.39%)	25.78 (↓ 23.08%)	22.16 (↓ 10.51%)
Case3	Track-1	29.96	64.87 (↓ 53.82%)	58.22 (↓ 48.54%)	35.83 (↓ 16.38%)
	Track-2	33.77	55.56 (↓ 39.22%)	51.58 (↓ 34.53%)	42.75 (↓ 21.01%)
	Track-3	41.24	62.69 (↓ 34.53%)	61.69 (↓ 33.15%)	48.01 (↓ 14.10%)

TABLE VI

MRES OF THE POSITION TRACKING FOR ALL TRAJECTORIES WITH GL-BASED ESTIMATION, GRNN-BASED ESTIMATION, UKF-BASED ESTIMATION, AND PROPOSED ALGORITHM, RESPECTIVELY

	Tracks	GL (m)	UKF (m)	GRNN (m)	Proposed method(m)
Case1	Track-1	0.007	0.006	0.005	0.004
	Track-2	0.006	0.006	0.005	0.004
	Track-3	0.007	0.007	0.006	0.005
Case2	Track-1	0.012	0.011	0.010	0.007
	Track-2	0.011	0.010	0.008	0.007
	Track-3	0.014	0.012	0.011	0.009
Case3	Track-1	0.029	0.028	0.016	0.014
	Track-2	0.028	0.026	0.021	0.017
	Track-3	0.030	0.029	0.024	0.020

TABLE VII

MEANS OF THE POSITION RMSE AND MRES OF THE POSITION TRACKING FOR THE TRACK IN THE FIELD EXPERIMENT

Method	GL	UKF	GRNN	Proposed method
RMSE (m)	37.23	35.71	26.92	19.44
MRE	0.043	0.038	0.035	0.023

measurement and the dimension of the state by d_m and d_n , respectively. Table VIII summarizes the model complexity of the proposed algorithm. The main procedures include CCGA optimization $O(n\text{Max}_{\text{iter}})$, pattern layer $O(Nd_m)$, and summation layer $O(Nd_n)$. Thus, the total model complexity is

$$\text{FLOPs}_{\text{model}} = (N + 1)d_n + Nd_m + n\text{Max}_{\text{iter}}. \quad (30)$$

The model complexity is on the order of $O(d_n)$ and $O(d_m)$, which has a 1st-order relationship with both the state dimension and the measurement dimension. It is also related to the trained sample size of CCGA-GRNN and the iteration number of CCGA. In practice, N and $n\text{Max}_{\text{iter}}$ are determined by the complexity of the motions of the targets of interest and the required estimation accuracy.

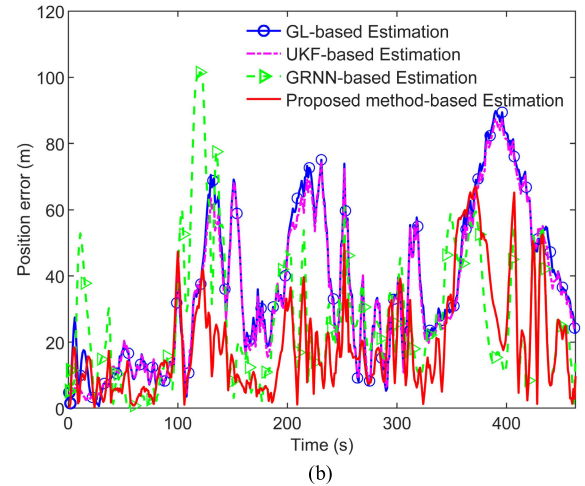
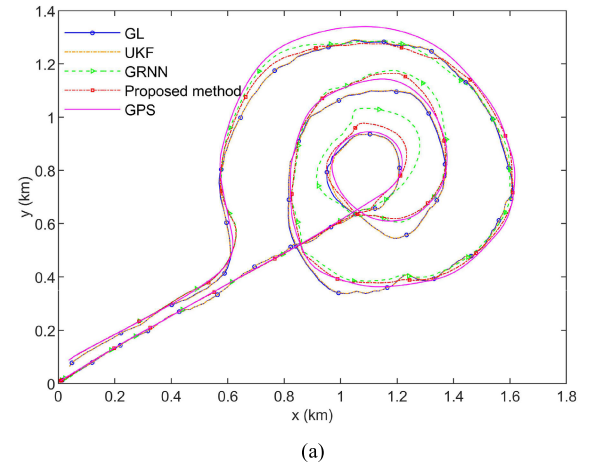


Fig. 13. Results of tracking in the Cartesian coordinates (real data). (a) Tracking results. (b) Position deviation of tracking track.

Nevertheless, the computational efficiency can be evaluated for a trained CCGA-GRNN. We test the computational time of the proposed algorithm, GL-based tracking method, and UKF by means of 100 experiment runs. For a fair comparison,

TABLE VIII

MODEL COMPLEXITY OF THE PROPOSED ALGORITHM IN TERMS OF MATRIX MULTIPLICATION

Processing Step	Multiplications
CCGA optimization	$S_{11} = nMax_{iter}$
Pattern layer $W_1^T z_k$	$S_{12} = Nd_m$
Summation layer $W_2^T h_{i,k}$	$S_{13} = Nd_n$
Output layer $\hat{t}_k$	$S_{14} = d_n$
Total model Complexity: $FLOPs_{model} = S_{11} + S_{12} + S_{13} + S_{14}$	

all the algorithms are tested by the same Intel Core i7-4790 CPU at 4.0 GHz and 32.0-GB RAM. Test results show that the proposed algorithm, GL, and UKF consume 39.80, 21.66, and 102.28 ms, respectively. Hence, the proposed algorithm is suitable for real-time implementation.

V. CONCLUSION

In this article, an efficient tracking algorithm for passive bistatic radar by exploiting the learning and nonlinear regression capabilities of GRNN is proposed. First, the GRNN method is advocated for handling the nonlinear relationship between the target state and the measurements. The study demonstrates that nonlinear function approximation from the target measurement to the target position is possible using the GRNN technique. Second, the CCGA technique is used for optimizing the sample size and smoothing factor of the GRNN model, which can further improve the localization performance. Furthermore, the KF framework is applied to the CCGA-GRNN model, which proves effective in forming precise trajectories. The simulations with various conditions of measurements scenario are conducted. These show that the comparative advantage of the proposed algorithm is more significant when the measurement noise is larger. Moreover, the proposed algorithm exhibits better performance in terms of RMSE and MRE compared with other state-of-the-art methods. The agreement between the tracking results and ground truth verifies the feasibility of the proposed algorithm via the field experiment. The position RMSE of the proposed algorithm for a passive bistatic system is about 19.44 m.

In the proposed algorithm, target association is performed before localization. The whole target association module is implemented in bistatic coordinates. In future work, the converted measurements from CCGA-GRNN are considered to conduct target association in the Cartesian coordinates. In complex detection environment, the influence of different measurement models on target tracking performance is worthy of research. In addition, target tracking of end-to-end using the mix-machine-learning techniques is another interesting topic.

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